MULTIPLE CHOICE

1. Which equation is a reciprocal identity?

A. $\sec\theta = \frac{1}{\tan\theta}$ **C.** $\sec\theta = \frac{1}{\csc\theta}$

B. $\sec\theta = \frac{1}{\cos\theta}$ **D.** $\sec\theta = \frac{1}{\sin\theta}$

ANS: B DIF: Easy OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities KEY: reciprocal identity

2. Which equation is a quotient identity?

A. $\cot\theta = \frac{1}{\cos\theta}$ **C.** $\cot\theta = \frac{\cos\theta}{\sin\theta}$

B. $\cot\theta = \frac{\sin\theta}{\cos\theta}$ **D.** $\cot\theta = \frac{1}{\sin\theta}$

ANS: C DIF: Easy OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities KEY: quotient identity

3. Which equation is a Pythagorean identity?

A. $1 + \tan^2\theta = \sec^2\theta$ **C.** $1 - \tan^2\theta = \sec^2\theta$

B. $\cot^2\theta - 1 = \csc^2\theta$ **D.** $\sin^2\theta - \cos^2\theta = 1$

ANS: A DIF: Easy OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities KEY: Pythagorean identities

4. What does the expression $\cot\theta\sin\theta\csc\theta$ simplify to?

A. $\cot\theta$ **C.** $\cos\theta$

B. $\tan\theta$ **D.** $\sin\theta$

ANS: A DIF: Average OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities
KEY: quotient identity | reciprocal identity

5. Which of the following sets is the set of reciprocal trigonometric ratios?

A. csc θ, sec θ, and cot θ **C.** sin θ, csc θ, and cot θ

B. sec θ, cos θ, and cot θ **D.** csc θ, sec θ, and tan θ

ANS: A DIF: Average OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities
KEY: reciprocal trigonometric ratios

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6. What is an equivalent expression for $\cot A$?

A. $\frac{1}{\sin A}$ **C.** $\frac{1}{\cos A}$

B. $\frac{1}{\tan A}$ **D.** $\frac{1}{\sec A}$

ANS: B DIF: Easy OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities
KEY: reciprocal trigonometric ratios

7. Which expression is equivalent to $\frac{\cos\theta}{1+\sin\theta}$?

A. $\frac{\cos\theta(1+\sin\theta)}{1+\sin^2\theta}$ **C.** $\frac{1-\sin\theta}{\cos\theta}$

B. $\frac{\cos\theta}{1-\sin\theta}$ **D.** $\frac{1+\sin\theta}{\cos\theta}$

ANS: C DIF: Average OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities KEY: trigonometric identity

8. Which expression is equivalent to $\tan\theta+\cot\theta$?

A. 1 **C.** $\frac{1}{\cos\theta\sin\theta}$

B. $\cos\theta\sin\theta$ **D.** $2\left(\frac{\sin\theta}{\cos\theta}\right)$

ANS: C DIF: Average OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities KEY: trigonometric identity

9. Which expression is equivalent to $\sec^2\theta\sin^2\theta$?

A. $\tan^2\theta$ **C.** 1

B. $\cos^2\theta$ **D.** $\sin^4\theta$

ANS: A DIF: Easy OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities KEY: trigonometric identity

10. Which expression is equivalent to $\tan^2\theta-\sin^2\theta$?

A. $\sin^2\theta\cos^2\theta$ **C.** $\frac{\sin^3\theta}{\cos^2\theta}$

B. $\sin^2\theta\tan^2\theta$ **D.** $\frac{\sin^2\theta}{\cos^2\theta}$

ANS: B DIF: Average OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities KEY: trigonometric identity

11. Which expression is equivalent to $(\sin\theta+\cos\theta)^2$?

A. $1+2\sin\theta\cos\theta$ **C.** 1

B. $2\sin\theta\cos\theta$ **D.** $\sin^2\theta+\cos^2\theta$

ANS: A DIF: Average OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities KEY: trigonometric identity

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12. Which expression is equivalent to $1-2\sin^2\theta$?

A. $\cos^2\theta$ **C.** $1-2\cos^2\theta$

B. $2\cos^2\theta-1$ **D.** $\sin^2\theta$

ANS: B DIF: Average OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities KEY: trigonometric identity

13. Express $\left(\frac{\cos\alpha-\sec\alpha}{\sec\alpha}+\cos^2\alpha\tan^2\alpha\right)\left(\frac{\tan\alpha-\sin\alpha}{\tan\alpha}\right)$ in terms of cosine and sine only, and simplify.

A. 0 **C.** 1

B. $\cos^2\alpha$ **D.** $\sin^2\alpha$

ANS: A DIF: Difficult OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities KEY: trigonometric identity

14. Which expression is equivalent to $\cos x\cos y-\sin x\sin y$?

A. $\sin(x+y)$ **C.** $\sin(x-y)$

B. $\cos(x+y)$ **D.** $\cos(x-y)$

ANS: B DIF: Average OBJ: Section 6.2 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities KEY: sum identities

15. Which expression is equivalent to $\frac{\tan A-\tan B}{1+\tan A\tan B}$?

A. $\tan(A+B)$ **C.** $\tan(A-B)$

B. $\cot(A+B)$ **D.** $\cot(A-B)$

ANS: C DIF: Average OBJ: Section 6.2 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities
KEY: tangent | sum identities | difference identities

16. Determine the exact value of $\cos 15°$.

A. $2\sqrt{2}$ **C.** $\frac{\sqrt{6}-\sqrt{2}}{4}$

B. $\frac{2\sqrt{2}}{3}$ **D.** $\frac{\sqrt{6}+\sqrt{2}}{4}$

ANS: D DIF: Average OBJ: Section 6.2 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities KEY: difference identities

17. Simplify $\cos 19°\cos 161°-\sin 19°\sin 161°$.

A. 0 **C.** –1

B. undefined **D.** 1

ANS: C DIF: Easy OBJ: Section 6.2 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities KEY: sum identities | evaluate

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18. Simplify $\sin 168° \cos 143° - \cos 168° \sin 143°$. Round your answer to the nearest hundredth.

A. 0.47
C. –0.75
B. –1.15
D. 0.42

ANS: D DIF: Average OBJ: Section 6.2 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities
KEY: sum identities | difference identities | evaluate

19. Simplify $\dfrac{\tan(116°) + \tan(-26°)}{1 - \tan(116°)\tan(-26°)}$.

A. –1
C. 0
B. 1
D. undefined

ANS: D DIF: Easy OBJ: Section 6.2 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities KEY: sum identities | evaluate

20. Which expression is equivalent to $\dfrac{\cos(A+B)}{\cos(A-B)}$?

A. $\dfrac{1+\tan A \tan B}{1-\tan A \tan B}$
C. $\cos A \sin B$
B. $\dfrac{1-\tan A \tan B}{1+\tan A \tan B}$
D. $\sin A \cos B$

ANS: B DIF: Difficult OBJ: Section 6.2 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities
KEY: sum identities | evaluate | difference identities

21. Which expression is equivalent to $\dfrac{\sin(2B)}{2\sin^2 B}$?

A. $2\csc B$
C. $\tan B$
B. $\dfrac{2}{\sin B}$
D. $\cot B$

ANS: D DIF: Average OBJ: Section 6.2 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities KEY: double-angle identities

22. $\dfrac{\sin 4\theta}{1+\cos 4\theta}$ is equivalent to

A. $\tan 2\theta$
C. $\tan \theta$
B. $\tan^2 \theta$
D. $2\tan \theta$

ANS: A DIF: Difficult OBJ: Section 6.2 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities KEY: double-angle identities

23. $4\cos^3 \theta - 3\cos \theta$ can be rewritten as

A. $\cos 3\theta$
C. $\cos \theta$
B. $3\cos \theta$
D. $\cos^2 \theta$

ANS: A DIF: Difficult OBJ: Section 6.2 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities
KEY: double-angle identities | sum identities

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24. $\frac{1-\cos x}{\sin x}$ can be rewritten as

A. $2\tan x$

B. $\frac{\tan x}{2}$

C. $2\tan\frac{x}{4}$

D. $\tan\frac{x}{2}$

ANS: D DIF: Difficult + OBJ: Section 6.2 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities KEY: half-angle identities

25. What is the solution to the equation $\sin x + 4\csc x + 5 = 0$ over the domain $0 \le x \le 2\pi$?

A. $\frac{3\pi}{4}$

B. $\frac{\pi}{2}$

C. $\frac{\pi}{4}$

D. $\frac{3\pi}{2}$

ANS: D DIF: Average OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities KEY: reciprocal identity

26. What is the solution to the equation $\tan^2 x + 3\sec x + 3 = 0$ over the domain $0° \le x \le 360°$?

A. 60° and 240°

B. 120°, 180°, and 240°

C. no solution

D. 120°

ANS: B DIF: Average OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities KEY: reciprocal identity

27. What is the general solution, in degress, to the equation $2\cos x \cos 2x - 2\sin x \sin 2x = -1$?

A. $40° + 180n°$ and $80° + 180n°$. where $n \in \text{I}$

B. $40° + 120n°$, where $n \in \text{I}$

C. $40° + 120n°$ and $80° + 120n°$, where $n \in \text{I}$

D. $220° + 120n°$ and $330° + 120n°$, where $n \in \text{I}$

ANS: C DIF: Difficult OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities
KEY: double-angle identities | general solutions

28. What is the general solution, in radians, to the equation $(4\cos^2 2\theta + 1)\sin\frac{1}{3}\theta = 0$?

A. $2\pi n$ where $n \in \text{I}$

B. no solution

C. $3\pi n$ where $n \in \text{I}$

D. $\frac{\pi}{3}n$ where $n \in \text{I}$

ANS: C DIF: Difficult OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities
KEY: double-angle identities | general solutions

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29. What is the general solution, in radians, to the equation $\sin 2x + 2\cos x = 0$?

A. $\pi + \frac{\pi}{2}n$, where $n \in \mathrm{I}$

B. $2\pi n$, where $n \in \mathrm{I}$

C. no solution

D. $\frac{\pi}{2} + \pi n$, where $n \in \mathrm{I}$

ANS: D DIF: Average OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities
KEY: double-angle identities | general solutions

COMPLETION

1. $\tan x = \frac{\sin x}{\cos x}$ and $\cot x = \frac{\cos x}{\sin x}$ are known as the _____ identities.

ANS: quotient

DIF: Easy OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities
KEY: reciprocal identities | quotient identities | Pythagorean identities

2. An equation with trigonometric expressions that is true for all values of the variable for which the expressions are defined is called a trigonometric _____ .

ANS: identity

DIF: Easy OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities KEY: trigonometric identity

3. The general solution, in degrees, for the equation $\cos x - \frac{1}{2} = 0$ is _____ .

ANS: $60° + 360n°$ and $300° + 360n°$, where $n \in \mathrm{I}$

DIF: Average OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities KEY: general solutions

4. The reciprocal identities are defined as $\csc x =$ _____,

$\sec x =$ _____, and

$\cot x =$ _____

ANS: $\frac{1}{\sin x}, \frac{1}{\cos x}, \frac{1}{\tan x}$

DIF: Easy OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities
KEY: reciprocal identities | quotient identities | Pythagorean identities

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5. A trigonometric identity is a relation among trigonometric ratios that is _____ for all values of the variables for which both sides are _____.

ANS: true, defined

DIF: Average OBJ: Section 6.3 NAT: T6 TOP: Proving Identities
KEY: trigonometric identity

MATCHING

Match the corresponding trigonometric expressions to form trigonometric identities.

A. $\frac{\sin(\alpha+\beta)}{\cos(\alpha+\beta)}$

B. $\frac{1-\tan\alpha\tan\beta}{1+\tan\alpha\tan\beta}$

C. $\frac{\csc\alpha\cos\alpha}{\sec\alpha\sin\alpha}$

D. $\tan\alpha$

E. $\tan\alpha-\sec\alpha$

1. $\frac{\sin\alpha-1}{\cos\alpha}$

2. $(\csc\alpha-1)(1+\csc\alpha)$

3. $\frac{\sin\alpha+\cos\alpha\tan\beta}{\cos\alpha-\sin\alpha\tan\beta}$

4. $\frac{\sin 4\alpha-\sin 2\alpha}{\cos 2\alpha+\cos 4\alpha}$

5. $\frac{\cos(\alpha+\beta)}{\cos(\alpha-\beta)}$

1. ANS: E DIF: Average OBJ: Section 6.3 NAT: T6
TOP: Proving Identities KEY: quotient identities | reciprocal identities

2. ANS: C DIF: Average OBJ: Section 6.3 NAT: T6
TOP: Proving Identities KEY: reciprocal identities

3. ANS: A DIF: Average OBJ: Section 6.3 NAT: T6
TOP: Proving Identities KEY: difference identities | sum identities

4. ANS: D DIF: Difficult + OBJ: Section 6.3 NAT: T6
TOP: Proving Identities KEY: double-angle identities

5. ANS: B DIF: Average OBJ: Section 6.3 NAT: T6
TOP: Proving Identities KEY: difference identities | sum identities

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SHORT ANSWER

1. Find the exact value of $\cos\frac{2\pi}{9}\cos\frac{\pi}{18}+\sin\frac{2\pi}{9}\sin\frac{\pi}{18}$.

ANS:

$$\begin{aligned}\cos\frac{2\pi}{9}\cos\frac{\pi}{18}+\sin\frac{2\pi}{9}\sin\frac{\pi}{18} &= \cos\left(\frac{2\pi}{9}-\frac{\pi}{18}\right)\\ &= \cos\left(\frac{4\pi}{18}-\frac{\pi}{18}\right)\\ &= \cos\left(\frac{3\pi}{18}\right)\\ &= \cos\left(\frac{\pi}{6}\right)\\ &= \frac{\sqrt{3}}{2}\end{aligned}$$

DIF: Average OBJ: Section 6.2 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities
KEY: difference identities | exact value

2. Find the exact value of $\dfrac{\tan\frac{\pi}{10}+\tan\frac{\pi}{15}}{1-\tan\frac{\pi}{10}\tan\frac{\pi}{15}}$.

ANS:

$$\begin{aligned}\frac{\tan\frac{\pi}{10}+\tan\frac{\pi}{15}}{1-\tan\frac{\pi}{10}\tan\frac{\pi}{15}} &= \tan\left(\frac{\pi}{10}+\frac{\pi}{15}\right)\\ &= \tan\left(\frac{3\pi}{30}+\frac{2\pi}{30}\right)\\ &= \tan\left(\frac{5\pi}{30}\right)\\ &= \tan\left(\frac{\pi}{6}\right)\\ &= \frac{1}{\sqrt{3}} \text{ or } \frac{\sqrt{3}}{3}\end{aligned}$$

DIF: Difficult OBJ: Section 6.2 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities KEY: sum identities | exact value

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3. Determine the value of $\cos\theta\left(\frac{\sin\theta\sec\theta+\tan\theta}{\sec\theta\tan\theta}\right)$ if $\sin\theta=\frac{13}{85}$ and $\cos\theta=\frac{84}{85}$.

ANS:
Simplify the expression.

$$\cos\theta\left(\frac{\sin\theta\sec\theta+\tan\theta}{\sec\theta\tan\theta}\right)=\cos\theta\left(\frac{\frac{\sin\theta}{\cos\theta}+\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos\theta}\,\frac{\sin\theta}{\cos\theta}}\right)$$

$$=\cos\theta\left(\frac{\frac{2\sin\theta}{\cos\theta}}{\frac{\sin\theta}{\cos^2\theta}}\right)$$

$$=\cos\theta\left(\frac{2\sin\theta}{\cos\theta}\right)\left(\frac{\cos^2\theta}{\sin\theta}\right)$$

$$=2\cos^2\theta$$

Substitute $\cos\theta=\frac{84}{85}$:

$$2\left(\frac{84}{85}\right)^2=\frac{14\,112}{7225}$$

Or, substitute directly. Since $\sin\theta=\frac{13}{85}$ and $\cos\theta=\frac{84}{85}$, then $\sec\theta=\frac{85}{84}$, and

$$\tan\theta=\frac{\sin\theta}{\cos\theta}$$

$$=\frac{\frac{13}{85}}{\frac{84}{85}}$$

$$=\frac{13}{84}$$

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$$\cos\theta\left(\frac{\sin\theta\sec\theta+\tan\theta}{\sec\theta\tan\theta}\right)=\frac{84}{85}\left(\frac{\left(\frac{13}{85}\right)\left(\frac{85}{84}\right)+\frac{13}{84}}{\left(\frac{85}{84}\right)\left(\frac{13}{84}\right)}\right)$$

$$=\frac{84}{85}\left(\frac{\frac{13}{84}+\frac{13}{84}}{\frac{(85)(13)}{7056}}\right)$$

$$=\frac{84}{85}\left(\frac{\frac{26}{84}}{\frac{(85)(13)}{7056}}\right)$$

$$=\frac{84}{85}\left(\frac{26}{84}\right)\left(\frac{7056}{(85)(13)}\right)$$

$$=\frac{2(7056)}{85^2}$$

$$=\frac{14\,112}{7225}$$

DIF: Difficult OBJ: Section 6.1 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities KEY: exact value

4. Use a counterexample to show that $\cos(x+y)=\cos x+\cos y$ is not an identity.

ANS:
Answers may vary. Sample answer:

Use $x=0$ and $y=\frac{\pi}{2}$:

L.S. $=\cos(x+y)$

$=\cos\left(0+\frac{\pi}{2}\right)$

$=\cos\frac{\pi}{2}$

$=0$

R.S. $=\cos x+\cos y$

$=\cos 0+\cos\frac{\pi}{2}$

$=1+0$

$=1$

L.S. ≠ R.S.

Thus, $\cos(x+y)=\cos x+\cos y$ is not an identity.

DIF: Average OBJ: Section 6.3 NAT: T6 TOP: Proving Identities
KEY: counterexample

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5. Prove the identity $\tan^2\theta - \sin^2\theta = \sin^2\theta \tan^2\theta$.

ANS:

L.S. $= \tan^2\theta - \sin^2\theta$ R.S. $= \sin^2\theta \tan^2\theta$

$$= \frac{\sin^2\theta}{\cos^2\theta} - \sin^2\theta$$

$$= \frac{\sin^2\theta - \sin^2\theta\cos^2\theta}{\cos^2\theta}$$

$$= \frac{\sin^2\theta(1 - \cos^2\theta)}{\cos^2\theta}$$

$$= \frac{\sin^2\theta(\sin^2\theta)}{\cos^2\theta}$$

$$= \sin^2\theta \tan^2\theta$$

L.S. = R.S.

DIF: Average OBJ: Section 6.1 | Section 6.3 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities | Proving Identities
KEY: Pythagorean identities | trigonometric identity

6. Is the equation $\csc^2\theta = 1 + \cot^2\theta$ true for all values of θ?

ANS:

L.S. $= \csc^2\theta$ R.S. $= 1 + \cot^2\theta$

$$= 1 + \frac{\cos^2\theta}{\sin^2\theta}$$

$$= \frac{\sin^2\theta}{\sin^2\theta} + \frac{\cos^2\theta}{\sin^2\theta}$$

$$= \frac{\sin^2\theta + \cos^2\theta}{\sin^2\theta}$$

$$= \frac{1}{\sin^2\theta}$$

$$= \csc^2\theta$$

L.S. = R.S.

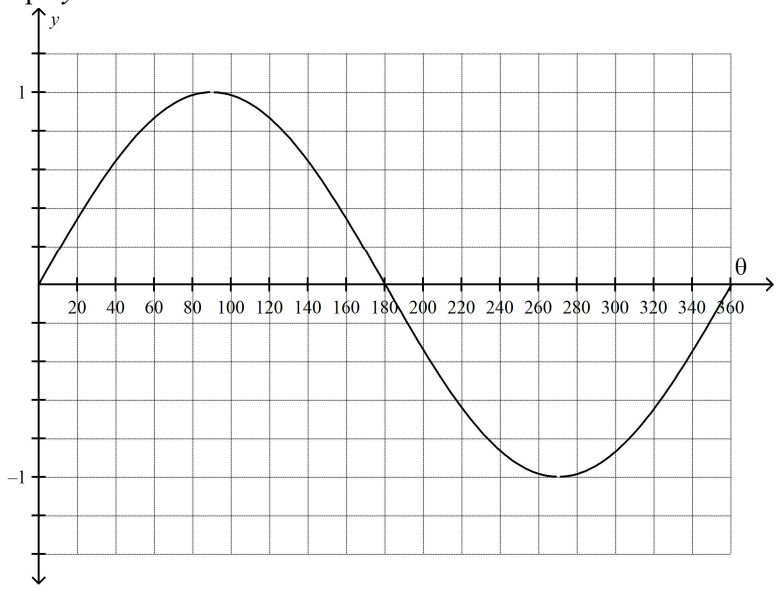
The identity is true for all values of θ.

DIF: Average OBJ: Section 6.1 | Section 6.3 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities | Proving Identities
KEY: Pythagorean identities | trigonometric identity

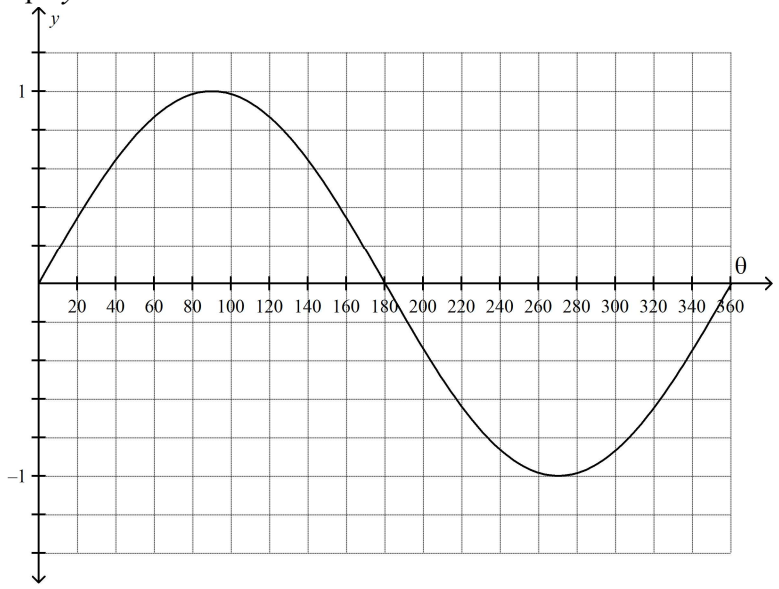
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7. Verify graphically that $\cos x \tan x = \sin x$ for $0° \le x \le 360°$.

ANS:
Graph $y = \cos x \tan x$



Graph $y = \sin x$



The graphs are the same for the range $0° \le x \le 360°$.

DIF: Easy OBJ: Section 6.3 NAT: T6 TOP: Proving Identities
KEY: reciprocal identities | trigonometric identity | graphing

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8. What is the solution for $2\cos x - \sqrt{3} = 0$ for $0 \le x \le 2\pi$?

ANS:

$$2\cos x - \sqrt{3} = 0$$

$$2\cos x = \sqrt{3}$$

$$\cos x = \frac{\sqrt{3}}{2}$$

$$x = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{\pi}{6}$$

Since cosine is also positive in quadrant IV, another solution is $2\pi - \frac{\pi}{6} = \frac{11\pi}{6}$.

DIF: Easy OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities KEY: exact value

9. Solve $\cot^2\theta + \cot\theta = 0$. State the solution in general form.

ANS:

$$\cot^2\theta + \cot\theta = 0$$

$$\cot\theta(\cot\theta + 1) = 0$$

$$\cot\theta = 0 \text{ or } \cot\theta = -1$$

$$\theta = \frac{\pi}{2} \text{ or } \theta = \frac{3\pi}{4}$$

Since the period for $\cot\theta$ is π, the solution in general form is $\theta = \frac{\pi}{2} + n\pi$ and $\theta = \frac{3\pi}{4} + n\pi$, where $n \in \mathrm{I}$.

DIF: Difficult OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities
KEY: exact value | general solutions

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10. Solve $\sec^2\theta - 2\tan\theta - 3 = 0$. State the general solution to the nearest degree.

ANS:

$$\sec^2\theta - 2\tan\theta - 3 = 0$$

$$(1 + \tan^2\theta) - 2\tan\theta - 3 = 0$$

$$\tan^2\theta - 2\tan\theta - 2 = 0$$

Use the quadratic formula.

$$\tan\theta = \frac{2 \pm \sqrt{12}}{2}$$

$$= 1 \pm \sqrt{3}$$

$$\approx 2.732 \text{ or } -0.732$$

$$\theta = \tan^{-1}(2.732) \text{ or } \tan^{-1}(-0.732)$$

$$\approx 70° \text{ or } -36°$$

Since the period for $\tan\theta$ is 180°, a positive solution corresponding to −36° is −36° + 180° or 144°. The general solution is $70° + 180°n$ and $144° + 180°n$, where $n \in \text{I}$.

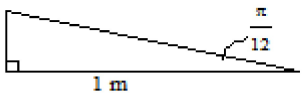
DIF: Difficult OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities
KEY: exact value | general solutions | quadratic formula

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PROBLEM

1. A skateboard ramp is built with an incline angle of $\frac{\pi}{12}$. If the base of the ramp is 1 m in length, determine the exact height of the ramp.

ANS:



From the triangle, $h = \tan\left(\frac{\pi}{12}\right)$.

$\frac{\pi}{12} = \frac{\pi}{3} - \frac{\pi}{4}$

Apply a compound angle formula for tangent.

$\tan\left(\frac{\pi}{12}\right) = \tan\left(\frac{\pi}{3} - \frac{\pi}{4}\right)$

$$= \frac{\tan\frac{\pi}{3} - \tan\frac{\pi}{4}}{1 + \tan\frac{\pi}{3}\tan\frac{\pi}{4}}$$

$$= \frac{\sqrt{3} - 1}{1 + \sqrt{3}}$$

The exact height of the ramp is $\frac{\sqrt{3} - 1}{1 + \sqrt{3}}$ m.

DIF: Difficult OBJ: Section 6.2 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities KEY: difference identities

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2. Prove the identity $1+\cos\theta=\dfrac{\sin^2\theta}{1-\cos\theta}$.

ANS:

L.S. $=1+\cos\theta$

$$\text{R.S.}=\frac{\sin^2\theta}{1-\cos\theta}$$

$$=\frac{1-\cos^2\theta}{1-\cos\theta}$$

$$=\frac{(1-\cos\theta)(1+\cos\theta)}{1-\cos\theta}$$

$$=1+\cos\theta$$

L.S. = R.S.

Therefore, $1+\cos\theta=\dfrac{\sin^2\theta}{1-\cos\theta}$.

DIF: Average OBJ: Section 6.1 | Section 6.3 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities | Proving Identities
KEY: Pythagorean identities | proof

3. Prove the identity $\sin\left(\dfrac{\pi}{2}-x\right)\cot\left(\dfrac{\pi}{2}+x\right)=-\sin x$.

ANS:

L.S. $=\sin\left(\dfrac{\pi}{2}-x\right)\cot\left(\dfrac{\pi}{2}+x\right)$ R.S. $=-\sin x$

$$=\cos x(-\tan x)$$

$$=\cos x\left(-\frac{\sin x}{\cos x}\right)$$

$$=-\sin x$$

L.S. = R.S.

Therefore, $\sin\left(\dfrac{\pi}{2}-x\right)\cot\left(\dfrac{\pi}{2}+x\right)=-\sin x$.

DIF: Average OBJ: Section 6.1 | Section 6.2 | Section 6.3
NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities | Sum, Difference, and Double-Angle Identities | Proving Identities
KEY: difference identities | sum identities | quotient identities | proof

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4. Prove the identity $\frac{\cos\theta - \sin\theta}{\cos\theta + \sin\theta} = \sec 2\theta - \tan 2\theta$.

ANS:

L.S. $= \frac{\cos\theta - \sin\theta}{\cos\theta + \sin\theta}$

R.S. $= \sec 2\theta - \tan 2\theta$

$$= \frac{1}{\cos 2\theta} - \frac{\sin 2\theta}{\cos 2\theta}$$

$$= \frac{1 - \sin 2\theta}{\cos 2\theta}$$

$$= \frac{1 - 2\sin\theta\cos\theta}{\cos^2\theta - \sin^2\theta}$$

$$= \frac{\sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta}{(\cos\theta - \sin\theta)(\cos\theta + \sin\theta)}$$

$$= \frac{(\cos\theta - \sin\theta)(\cos\theta - \sin\theta)}{(\cos\theta - \sin\theta)(\cos\theta + \sin\theta)}$$

$$= \frac{\cos\theta - \sin\theta}{\cos\theta + \sin\theta}$$

L.S. = R.S.

Therefore, $\frac{\cos\theta - \sin\theta}{\cos\theta + \sin\theta} = \sec 2\theta - \tan 2\theta$.

DIF: Difficult OBJ: Section 6.1 | Section 6.2 | Section 6.3
NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities | Sum, Difference, and Double-Angle Identities | Proving Identities KEY: Pythagorean identities | double-angle identities

5. Express $\sin(x + y + z)$ in terms of cosines and sines of x, y, and z.

ANS:

$$\sin(x + y + z) = \sin(x + y)\cos z + \cos(x + y)\sin z$$

$$= (\sin x\cos y + \cos x\sin y)\cos z + (\cos x\cos y - \sin x\sin y)\sin z$$

$$= \sin x\cos y\cos z + \cos x\sin y\cos z + \cos x\cos y\sin z - \sin x\sin y\sin z$$

DIF: Difficult + OBJ: Section 6.2 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities KEY: sum identities

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6. If $\sin x + \cos x = k$, for what value(s) of k does $\sin x \cos x = 1$?

ANS:

$$\sin x + \cos x = k$$

$$(\sin x + \cos x)^2 = k^2$$

$$\sin^2 x + 2\sin x \cos x + \cos^2 x = k^2$$

$$1 + 2\sin x \cos x = k^2$$

$$\sin x \cos x = \frac{k^2 - 1}{2}$$

Substitute $\sin x \cos x = 1$.

$$1 = \frac{k^2 - 1}{2}$$

$$2 = k^2 - 1$$

$$3 = k^2$$

$$k = \pm\sqrt{3}$$

DIF: Average OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities
KEY: Pythagorean identities | equation

7. Prove the identity $\frac{1 - \cos 2\theta + \sin 2\theta}{1 + \cos 2\theta + \sin 2\theta} = \tan\theta$.

ANS:

L.S. $= \frac{1 - \cos 2\theta + \sin 2\theta}{1 + \cos 2\theta + \sin 2\theta}$ L.S. $= \tan\theta$

$$= \frac{1 - (1 - 2\sin^2\theta) + 2\sin\theta\cos\theta}{1 + (2\cos^2\theta - 1) + 2\sin\theta\cos\theta}$$

$$= \frac{2\sin^2\theta + 2\sin\theta\cos\theta}{2\cos^2\theta + 2\sin\theta\cos\theta}$$

$$= \frac{2\sin\theta(\sin\theta + \cos\theta)}{2\cos\theta(\sin\theta + \cos\theta)}$$

$$= \frac{\sin\theta}{\cos\theta}$$

$$= \tan\theta$$

L.S. = R.S

Therefore, $\frac{1 - \cos 2\theta + \sin 2\theta}{1 + \cos 2\theta + \sin 2\theta} = \tan\theta$.

DIF: Average OBJ: Section 6.2 | Section 6.3 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities | Proving Identities
KEY: double-angle identities | proof

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8. Prove the identity $\frac{\cos^2\theta - \sin^2\theta}{\cos^2\theta + \sin\theta\cos\theta} = 1 - \tan\theta$.

ANS:

$$\text{L.S.} = \frac{\cos^2\theta - \sin^2\theta}{\cos^2\theta + \sin\theta\cos\theta}$$

$$= \frac{(\cos\theta - \sin\theta)(\cos\theta + \sin\theta)}{\cos\theta(\cos\theta + \sin\theta)}$$

$$= \frac{\cos\theta - \sin\theta}{\cos\theta}$$

$$\text{R.S.} = 1 - \tan\theta$$

$$= 1 - \frac{\sin\theta}{\cos\theta}$$

$$= \frac{\cos\theta - \sin\theta}{\cos\theta}$$

L.S. = R.S.

DIF: Average OBJ: Section 6.3 NAT: T6 TOP: Proving Identities
KEY: trigonometric identity | proof

9. Prove the identity $\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = \frac{2}{\sin^2\theta}$.

ANS:

$$\text{L.S.} = \frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta}$$

$$= \frac{1 - \cos\theta + 1 + \cos\theta}{(1+\cos\theta)(1-\cos\theta)}$$

$$= \frac{2}{1 - \cos^2\theta}$$

$$= \frac{2}{\sin^2\theta}$$

$$\text{R.S.} = \frac{2}{\sin^2\theta}$$

L.S. = R.S.

DIF: Average OBJ: Section 6.1 | Section 6.3 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities | Proving Identities
KEY: Pythagorean identities | proof

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10. Prove the identity $\dfrac{1+\tan\theta}{1+\cot\theta} = \dfrac{1-\tan\theta}{\cot\theta - 1}$.

ANS:

L.S. $= \dfrac{1+\tan\theta}{1+\cot\theta}$

$= \dfrac{1+\dfrac{\sin\theta}{\cos\theta}}{1+\dfrac{\cos\theta}{\sin\theta}}$

$= \dfrac{\dfrac{\cos\theta+\sin\theta}{\cos\theta}}{\dfrac{\sin\theta+\cos\theta}{\sin\theta}}$

$= \left(\dfrac{\cos\theta+\sin\theta}{\cos\theta}\right)\left(\dfrac{\sin\theta}{\cos\theta+\sin\theta}\right)$

$= \dfrac{\sin\theta}{\cos\theta}$

R.S. $= \dfrac{1-\tan\theta}{\cot\theta - 1}$

$= \dfrac{1-\dfrac{\sin\theta}{\cos\theta}}{\dfrac{\cos\theta}{\sin\theta}-1}$

$= \dfrac{\dfrac{\cos\theta-\sin\theta}{\cos\theta}}{\dfrac{\cos\theta-\sin\theta}{\sin\theta}}$

$= \left(\dfrac{\cos\theta-\sin\theta}{\cos\theta}\right)\left(\dfrac{\sin\theta}{\cos\theta-\sin\theta}\right)$

$= \dfrac{\sin\theta}{\cos\theta}$

L.S. = R.S.

DIF: Average OBJ: Section 6.1 | Section 6.3 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities | Proving Identities
KEY: quotient identities | proof

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11. An angle satisfies the relation $(\sec\theta)(\cot\theta) = 1$.

a) Use the definition of the reciprocal trigonometric ratios to express the left side of the relation in terms of the sine and/or cosine ratios.

b) Determine the value(s) for the angle. Do not use a calculator.

c) Verify your answer to part b) using a calculator.

d) Show your answer to part b) using a unit circle.

ANS:

a) $\sec\theta\cot\theta = \left(\frac{1}{\cos\theta}\right)\left(\frac{\cos\theta}{\sin\theta}\right)$

$= \frac{1}{\sin\theta}$

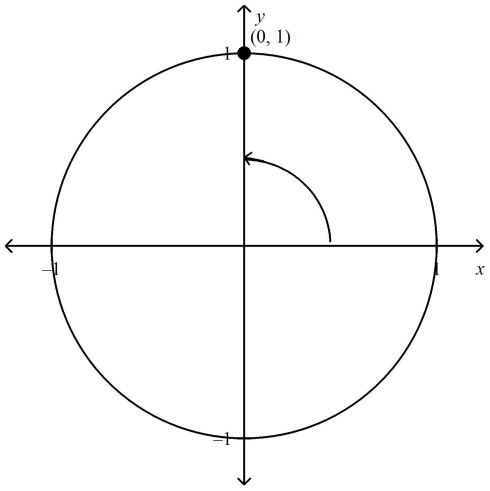
b) $\frac{1}{\sin\theta} = 1$

$1 = \sin\theta$

$\theta = 90°$

c) This is verified using a calculator.

d)



DIF: Average OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities
KEY: reciprocal trigonometric ratios | unit circle

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12. Use the expressions for the reciprocal trigonometric ratios in terms of x, y, and r to show that $\csc^2 A = 1 + \cot^2 A$ for all values of A.

ANS:

$\csc A = \frac{r}{y}, \cot A = \frac{x}{y}, x^2 + y^2 = r^2$

L.S. $= \csc^2 A$ R.S. $= 1 + \cot^2 A$

$= \left(\frac{r}{y}\right)^2$ $= 1 + \left(\frac{x}{y}\right)^2$

$= \frac{r^2}{y^2}$ $= 1 + \frac{x^2}{y^2}$

$= \frac{y^2 + x^2}{y^2}$

$= \frac{r^2}{y^2}$

L.S. = R.S.

This proves that the expression is valid for all values of A.

DIF: Average OBJ: Section 6.1 | Section 6.3 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities | Proving Identities
KEY: reciprocal trigonometric ratios | Pythagorean identities | proof

13. Prove that the equation $2\csc^2 x = \frac{1}{1+\cos x} + \frac{1}{1-\cos x}$ is true for all values of x.

ANS:

L.S. $= 2\csc^2 x$ R.S. $= \frac{1}{1+\cos x} + \frac{1}{1-\cos x}$

$= \frac{(1-\cos x) + (1+\cos x)}{(1+\cos x)(1-\cos x)}$

$= \frac{2}{1-\cos^2 x}$

$= \frac{2}{\sin^2 x}$

$= 2\csc^2 x$

L.S. = R.S.

This proves that the expression is valid for all values of x.

DIF: Average OBJ: Section 6.1 | Section 6.3 NAT: T6
TOP: Reciprocal, Quotient, and Pythagorean Identities | Proving Identities
KEY: reciprocal trigonometric ratios | proof

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14. Prove the identity $\frac{\sin 3\theta + \sin\theta}{\cos 3\theta + \cos\theta} = \tan 2\theta$.

ANS:

L.S. $= \frac{\sin 3\theta + \sin\theta}{\cos 3\theta + \cos\theta}$ R.S. $= \tan 2\theta$

$$= \frac{\sin 2\theta\cos\theta + \cos 2\theta\sin\theta + \sin\theta}{\cos 2\theta\cos\theta - \sin 2\theta\sin\theta + \cos\theta}$$

$$= \frac{2\sin\theta\cos^2\theta + (2\cos^2\theta - 1)\sin\theta + \sin\theta}{(2\cos^2\theta - 1)\cos\theta - 2\sin^2\theta\cos\theta + \cos\theta}$$

$$= \frac{\sin\theta}{\cos\theta}\left(\frac{2\cos^2\theta + 2\cos^2\theta - 1 + 1}{2\cos^2\theta - 1 - 2\sin^2\theta + 1}\right)$$

$$= \frac{\sin\theta}{\cos\theta}\left(\frac{4\cos^2\theta}{2\cos 2\theta}\right)$$

$$= \frac{4\sin\theta\cos^2\theta}{2\cos 2\theta\cos\theta}$$

$$= \frac{2\sin\theta\cos\theta}{\cos 2\theta}$$

$$= \frac{\sin 2\theta}{\cos 2\theta}$$

$$= \tan 2\theta$$

L.S. = R.S.

This proves that $\frac{\sin 3\theta + \sin\theta}{\cos 3\theta + \cos\theta} = \tan 2\theta$.

DIF: Difficult OBJ: Section 6.2 | Section 6.3 NAT: T6
TOP: Sum, Difference, and Double-Angle Identities | Proving Identities
KEY: sum identities | proof

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15. What is the solution to the equation $\cos 2x + 2 = \sin x$ for $0° \le x \le 360°$?

ANS:

$$\cos 2x + 2 = \sin x$$

$$1 - 2\sin^2 x + 2 = \sin x$$

$$2\sin^2 x + \sin x - 3 = 0$$

$$(2\sin x + 3)(\sin x - 1) = 0$$

$$\sin x = -\frac{3}{2} \text{ or } \sin x = 1$$

no solution or $x = 90°$

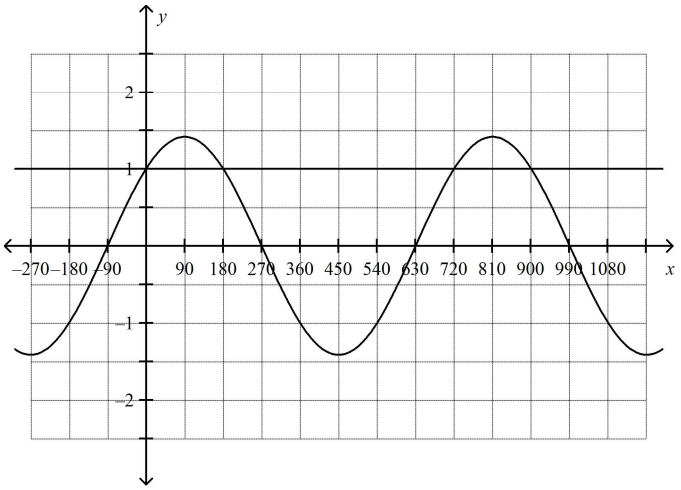
The solution is $x = 90°$.

DIF: Easy OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities
KEY: double-angle identities | equation

16. Solve $\cos\left(\frac{1}{2}x\right) + \sin\left(\frac{1}{2}x\right) = 1$.

ANS:

Graph the equations $y = 1$ and $y = \cos\left(\frac{1}{2}x\right) + \sin\left(\frac{1}{2}x\right)$ on the same set of axes and find the points of intersection.



From the graph, the solutions are 0°, 180°, 720°, 900°. The general solution is $x = 4n\pi$ or $x = (4n + 1)\pi$, where $n \in \mathrm{I}$.

DIF: Difficult + OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities KEY: general solutions | equation

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17. Solve $\sin 3x + \sin x = \cos x$ for $0 \le x \le 2\pi$.

ANS:

$$\sin 3x + \sin x = \cos x$$
$$\sin 2x \cos x + \cos 2x \sin x + \sin x = \cos x$$
$$2\sin x \cos^2 x + (2\cos^2 x - 1)\sin x + \sin x = \cos x$$
$$2\sin x \cos^2 x + 2\cos^2 x \sin x - \sin x + \sin x = \cos x$$
$$4\sin x \cos^2 x = \cos x$$
$$4\sin x \cos^2 x - \cos x = 0$$
$$\cos x(4\sin x \cos x - 1) = 0$$
$$\cos x(2\sin 2x - 1) = 0$$
$$\cos x = 0 \text{ or } \sin 2x = \frac{1}{2}$$
$$x = \frac{\pi}{2}, \frac{3\pi}{2} \text{ or } 2x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}$$
$$x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}$$

The solution is $x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}, \frac{\pi}{2}, \frac{3\pi}{2}$.

DIF: Difficult + OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities KEY: general solutions | equation

18. What is the general solution to $\tan x(\csc x + 2) = 0$?

ANS:

$\tan x(\csc x + 2) = 0$

$\tan x = 0$ $\quad$ $\csc x + 2 = 0$

$x = 0 + n\pi$ $\quad$ $\csc x = -2$

$= n\pi$ $\quad$ $\sin x = -\frac{1}{2}$

$x = -\frac{\pi}{6}$ or $x = \frac{11\pi}{6}$

$\sin x$ is also negative in quadrant III, so find a reflection of $x = \frac{11\pi}{6}$ in this quadrant. Another value for $\sin x = -\frac{1}{2}$ is $x = \frac{7\pi}{6}$.

Since the period of $\tan x$ is π and the period of $\sin x$ is 2π, the general solution is is $x = n\pi$ or $x = \frac{7\pi}{6} + n\pi$ or $x = \frac{11\pi}{6} + n\pi$, where $n \in \text{I}$.

DIF: Average OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities KEY: general solutions | equation

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19. Solve $4\sin^4 x + 3\sin^2 x - 1 = 0$ over the domain $0° \le x \le 360°$.

ANS:

$4\sin^4 x + 3\sin^2 x - 1 = 0$

$(4\sin^2 x - 1)(\sin^2 x + 1) = 0$ Divide both sides by $\sin^2 x + 1$ because it is always positive.

$4\sin^2 x - 1 = 0$

$\sin x = \pm\frac{1}{2}$

sin x is positive in quadrants I and II and negative in quadrants III and IV. The solution is $x = 30°, 150°, 210°, 330°$.

DIF: Average OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities
KEY: general solutions | quadratic equation

20. Solve $\sin^2 x = 0$ over the domain $0° \le x \le 360°$.

ANS:

$\sin^2 x = 0$

$\sin x = 0$

$x = 0°, 180°, 360°$

The solution is $x = 0°, 180°, 360°$.

DIF: Easy OBJ: Section 6.4 NAT: T6
TOP: Solving Trigonometric Equations Using Identities
KEY: general solutions | quadratic equation

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